

CARDIAC OUTPUT

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- **Definition** : Amount of blood pumped from each ventricle per unit time.
- Mainly refers to the **left ventricular output through the aorta**.
- Usually expressed in **3 ways**:
 - a. Stroke Volume
 - b. Minute Volume
 - c. Cardiac Index

EXPRESSIONS OF CARDIAC OUTPUT

- **1. Stroke Volume (SV) :—**
 - Amount of blood pumped out by each ventricle **during each beat**.
 - **Normal Value** : 70 mL (range: 60 – 80 mL) when Heart Rate (HR) is 72/min.
- **2. Minute Volume :—**
 - Amount of blood pumped out by each ventricle in **one minute**.
 - **Formula** :
$$\text{Minute Volume} = \text{Stroke Volume} \times \text{Heart Rate}$$
 - **Normal Value** : 5 L/ventricle/minute.
- **3. Cardiac Index :—**
 - Minute volume expressed in relation to **square meter (m^2) of body surface area**.
 - Amount of blood pumped out per ventricle / minute / square meter of body surface area.
 - **Normal Value** : $2.8 \pm 0.3 \text{ L}/m^2/\text{min}$.

Cardiac Reserve :—

Maximum amount of blood that can be pumped out by the heart above the normal resting value (*manifests as $\uparrow$ increasing cardiac output during heavy exercise*).

VARIATIONS IN CARDIAC OUTPUT

1. Physiological Variations

- a. **Age** : In children, cardiac output is ↓ lower than adults; in adults, it is ↑ higher.
- b. **Sex** : In females, cardiac output is less than males due to lower circulating blood volume.
- c. **Exercise** : ↑ Increases markedly due to an increase in both heart rate and force of contraction.
- d. **High Altitude** : ↑ Increases due to an ↑ increase in adrenaline secretion.
- e. **Hypoxia** : ↑ Increases cardiac output.
- f. **Posture** : Upright/erect posture ↓ decreases cardiac output due to pooling of blood in lower extremities.

2. Pathological Variations

Conditions with Increased (↑) Cardiac Output	Conditions with Decreased (↓) Cardiac Output
• Fever: Due to accelerated oxidative / metabolic processes • Anemia: Due to tissue hypoxia and compensatory vasodilation • Hyperthyroidism: Due to elevated Basal Metabolic Rate (BMR)	• Hypothyroidism: Due to decreased BMR • Shock: Due to poor pumping efficiency and circulation failure • Hemorrhage: Due to acute reduction in blood volume • Congestive Cardiac Failure (CCF): Weak myocardial contractions

FACTORS MAINTAINING / AFFECTING CARDIAC OUTPUT

Regulated by **4 primary determinants**:

- i. Venous return
- ii. Force of contraction
- iii. Heart rate
- iv. Peripheral resistance

1. VENOUS RETURN

- **Definition** : Amount of blood returned to the heart from different parts of the body.
- **Relationship** : Cardiac output is **directly proportional** to venous return.
- **5 Key Determinants of Venous Return** :—
- a. **Sympathetic Tone** :—

Sympathetic / Vasomotor Tone → Venoconstriction → Pushes blood towards heart → ↑ Venous Return

- b. **Gravity** :—
 - Gravitational pull reduces venous return by causing **venous pooling in legs** when standing upright → ↓ blood returning to heart.
- c. **Venous Pressure** :—
 - Normal pressure in venules is 12 – 18 mmHg.

- The pressure gradient along the venous tree serves as the driving force for venous return.

• d. **Respiratory Pump (Abdominothoracic Pump) :—**

- Respiratory movements facilitate venous return during **inspiration**.

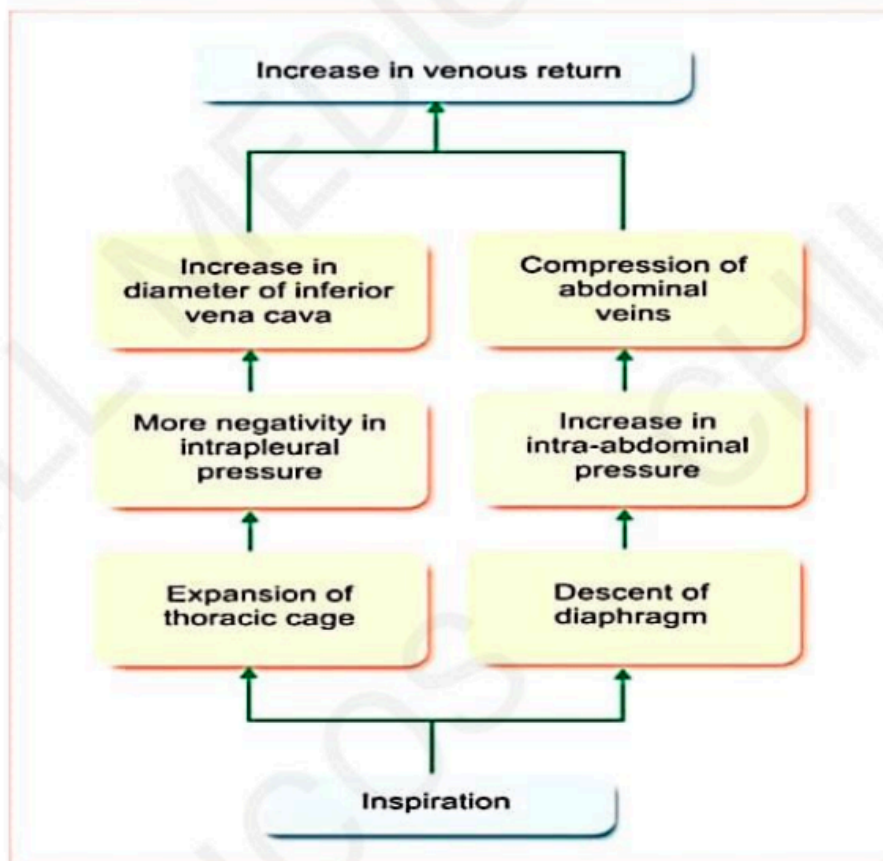
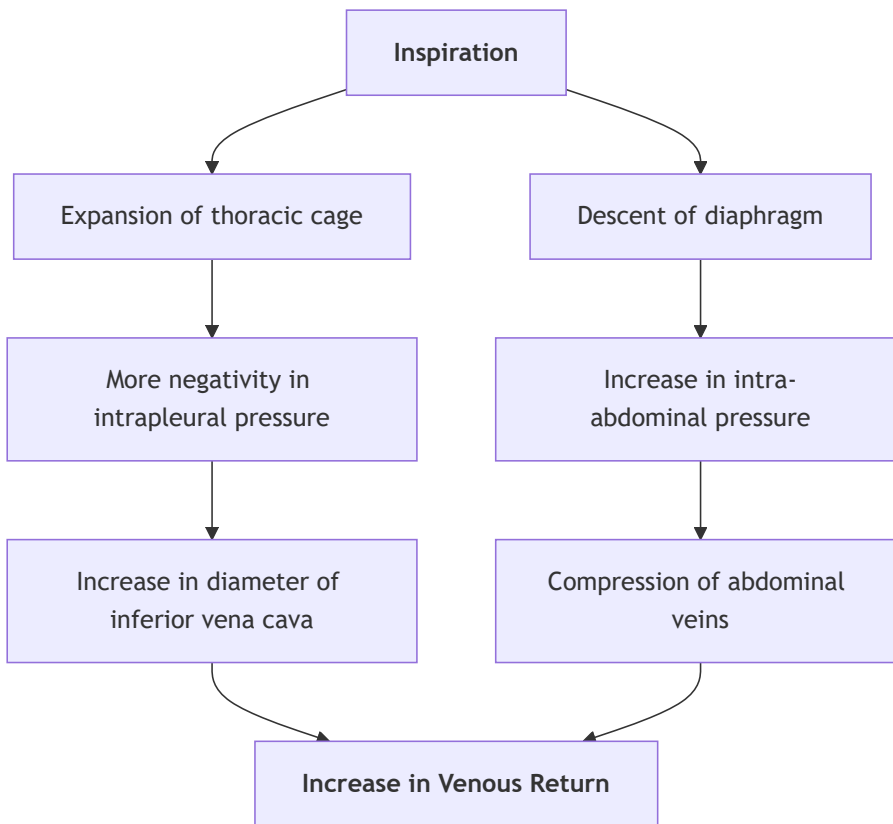
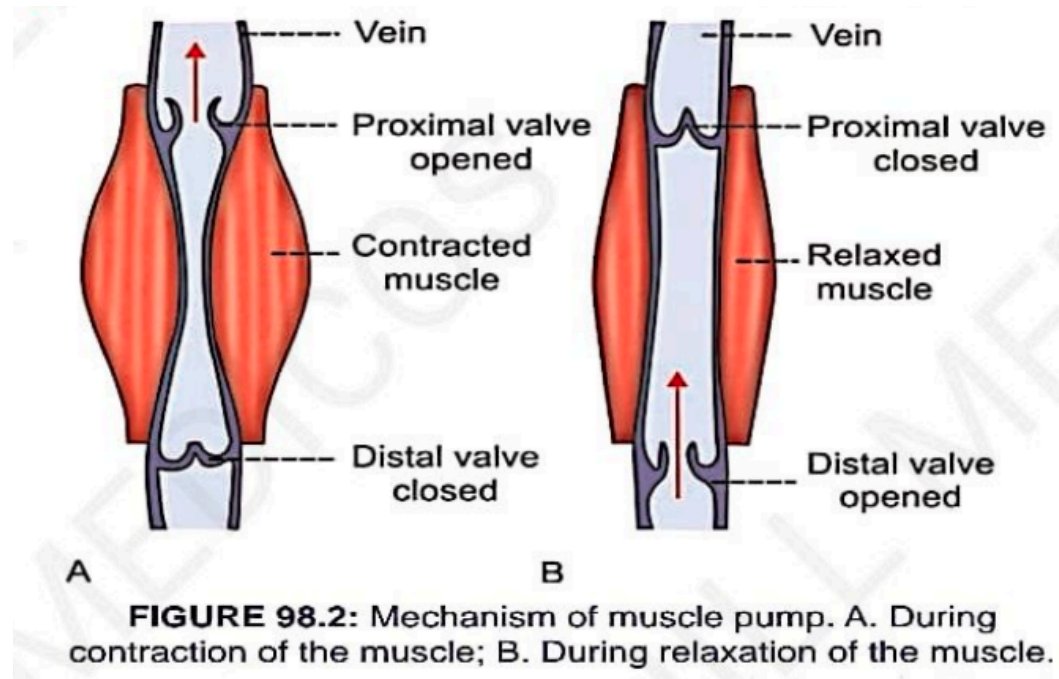


FIGURE 98.1: Effect of respiratory pump on venous return

- e. **Muscle Pump** :—
- During muscular contraction, deep veins are **compressed / squeezed**.
- Due to the presence of **one-way semilunar valves**, blood is propelled unidirectionally towards the heart.
- During muscle relaxation, proximal valves close to prevent backflow while distal valves open to allow venous refilling.



2. FORCE OF CONTRACTION (FOC)

- Cardiac output is **directly proportional** to the force of ventricular contraction.
- **Frank-Starling Law** :

Force of Contraction of Heart $\propto$ Initial Length of Muscle Fibers (End-Diastolic Volume)

- Force of contraction depends upon **Preload** and **Afterload**:

A. Preload:

- Stretching of cardiac muscle fibers at the **end of diastole**, just before contraction.
- Caused by increased intraventricular filling during diastole $\rightarrow \uparrow$ fiber length $\rightarrow \uparrow$ Force of Contraction $\rightarrow \uparrow$ Cardiac Output.
- **Directly proportional** to force of contraction and cardiac output.

B. Afterload:

- Force or resistance against which the ventricles must contract and eject blood (*primarily systemic vascular resistance / aortic pressure*).
- **Inversely proportional** to force of contraction and cardiac output.

3. HEART RATE

- Cardiac output is **directly proportional** to heart rate under physiological limits.

- **Note** : Moderate changes in heart rate do not significantly alter cardiac output due to compensatory adjustments in stroke volume.

4. PERIPHERAL RESISTANCE

- **Definition** : Resistance offered to blood flow at the peripheral blood vessels; the load against which the heart pumps blood.
- **Relationship** : Cardiac output is **inversely proportional** to peripheral resistance.
- **Resistant Vessels** : Resistance is mainly offered at the **arterioles**.
- **Site of Maximum Resistance** : Maximum peripheral resistance is offered in the **splanchnic vascular bed**.